EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

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Aim

To implement and explore the security, access-control, data-sharing and governance features of a Zoho Creator application.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use.

Application Used

The practical was carried out in Zoho Creator using the Job Application Form application. The application is used for managing job applications, and it served as the practical environment for exploring the security and governance features described in this record. The Zoho Creator account was on a trial (the footer shows “Trial expires in 4 days” with an Upgrade option), so some advanced features may depend on an upgrade; no upgrade-only feature is claimed as configured in this record.

Security and Governance Architecture

Figure 1 is a conceptual supporting diagram of the security and governance structure of the Job Application Form application. It is not a screenshot of the Zoho Creator interface.

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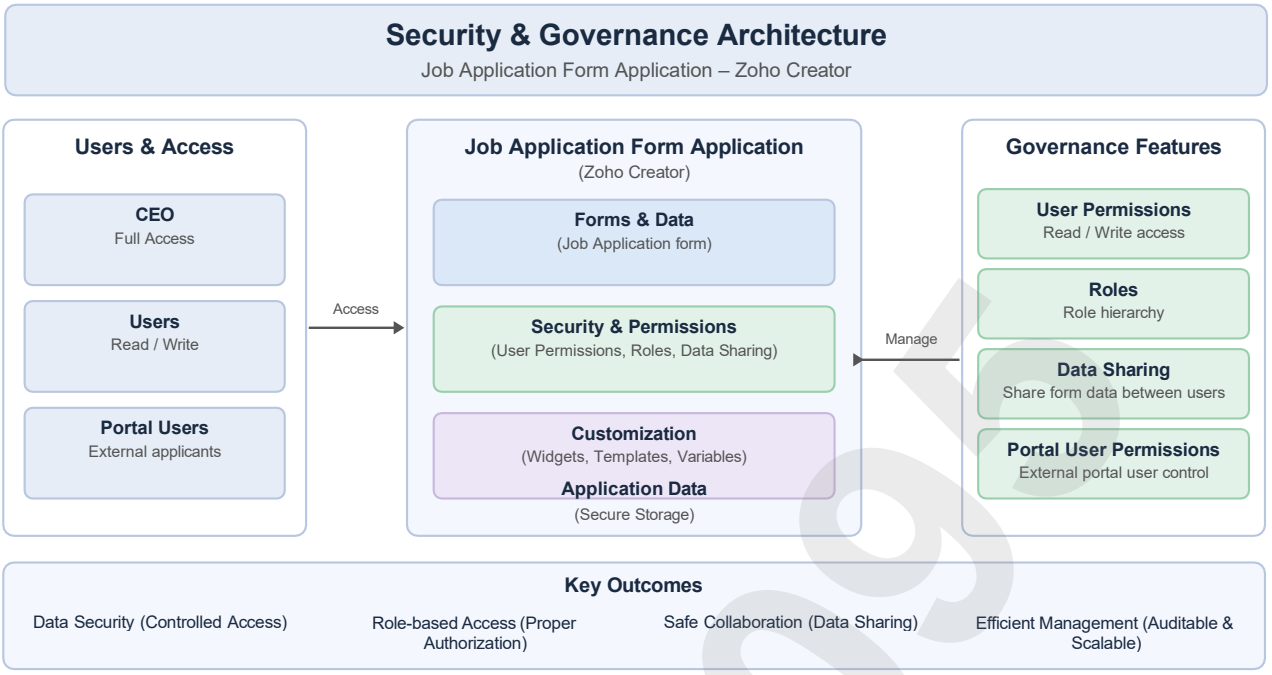


Figure 1: Security & Governance Architecture (Conceptual Diagram)

The diagram shows users (such as the CEO, users with Read / Write access and portal users) accessing the application, which contains forms and data, security and permissions (user permissions, roles and data sharing) and customisation items. The governance features on the right (user permissions, roles, data sharing and portal user permissions) manage and control access to the application. The key outcomes are data security through controlled access, role-based access, safe collaboration through data sharing, and efficient management.

User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records for each user. The Permissions tab lists the permission sets available in the application and provides an Add Permission option. The Permissions tab of the application listed three permission sets:

Name	Description (as displayed)
manager	supervising the event
sadhana	This profile will have read permission for all components
Write	This profile will have write permission for all components

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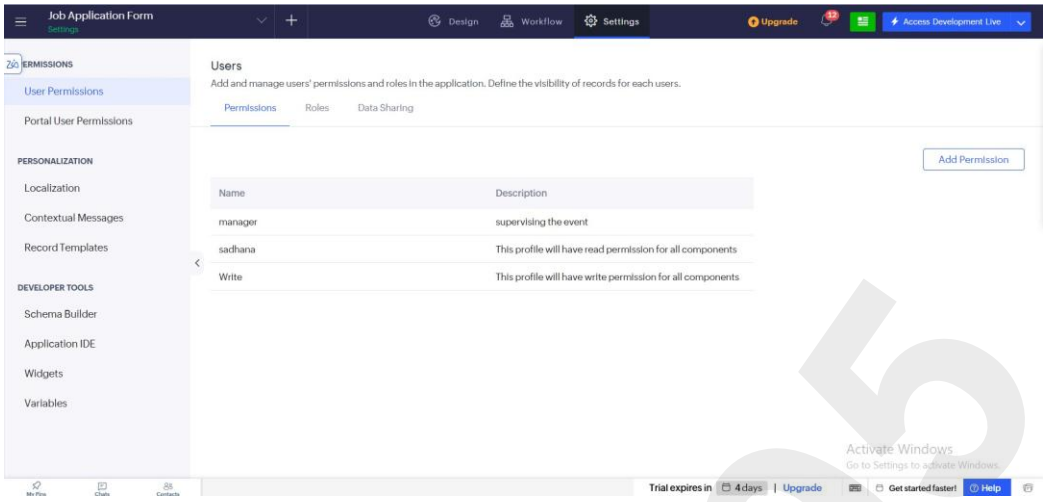


Figure 2: User Permissions in the Job Application Form application

A permission set with read permission allows a user to view the components and records of the application without changing them, whereas the Write permission set allows the user to make changes, such as adding or modifying records. The manager permission set is a custom set described as “supervising the event”.

Role-Based Access Control

The Roles tab shows the role hierarchy of the application. The default role CEO is displayed at the top, with the roles Manager and employee placed beneath it, and the options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available.

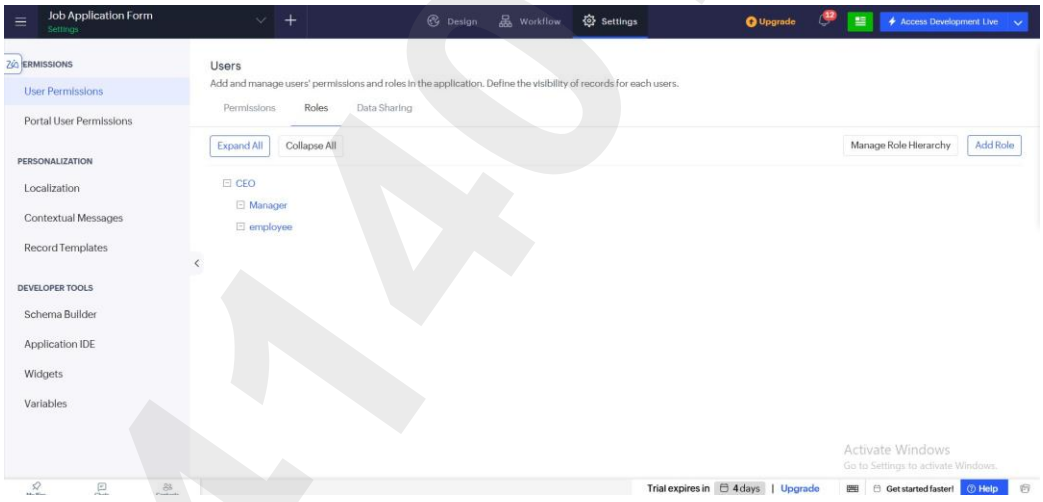


Figure 3: Roles and Role Hierarchy in the Job Application Form application

The role hierarchy of the Job Application Form application is summarised below:

Role	**Position in Hierarchy**
CEO	Top-level (default) role
Manager	Placed under CEO
employee	Placed under CEO

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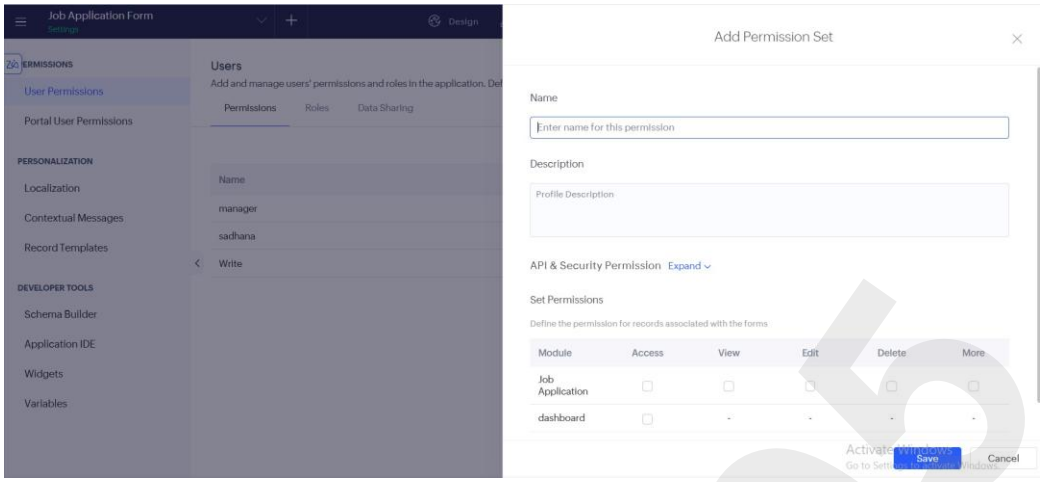


Figure 4: Add Permission Set Panel in the Job Application Form application

Because a user's role decides which records the user can see, the Manager and employee roles allow records to be viewed according to their position under the CEO. The Add Permission option opens the Add Permission Set panel shown in Figure 4. The panel asks for a Name and a Description, offers an expandable API & Security Permission section, and lists the modules Job Application and dashboard with the checkboxes Access, View, Edit, Delete and More under Set Permissions. The panel was opened to examine these options; no new permission set was created from it.

Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user from which records are shared, the role or user to which they are shared, and the level of access given. In this way, records can be made visible to additional roles or users beyond the default visibility of the role hierarchy. The Data Sharing tab of the application lists two data sharing rules, with the following details (R: denotes a role and U: denotes a user):

Figure 5: Data Sharing rules configured for the Job Application form

The saved rules appear in the Data Sharing list with the columns Name, Form, Shared From, Shared To and Level of access. The rule sadhana shares Job Application records of the Manager role with Read/Write

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access. The rule revathy shares Job Application records of the employee role with the user SHAMRUTI, who can only view them because the level of access is Read Only. The level of access determines how much a recipient can do with the shared records, so that visibility is extended in a controlled manner.

Security Features and Governance Controls

Figure 6 is a conceptual academic diagram that groups the security and governance controls of Zoho Creator. It is not a screenshot, and it is not proof that every listed feature was configured.

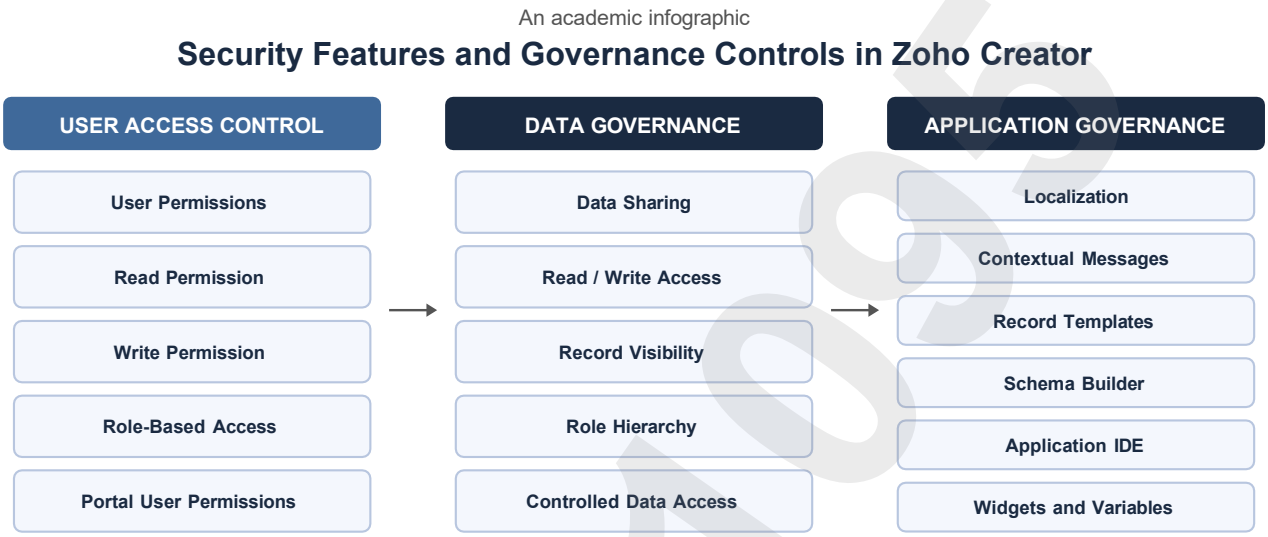


Figure 6: Security Features and Governance Controls in Zoho Creator (Conceptual Diagram)

Control	Explanation
User access control	User permissions (Read and Write), role-based access and portal user permissions determine who can use the application and what they can do.
Data governance	Data sharing, read / write access, record visibility and role hierarchy govern how application data is accessed.
Application governance	Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application.
Role hierarchy	Arranges roles in a reporting structure that influences which records a role can see.
Data sharing	Extends the visibility of records from one role or user to another with a chosen access level.
Record visibility	Decides which records are visible to which users, based on roles and sharing rules.
Administrative controls	Allow an administrator to manage permissions, roles and application settings.

In this practical, user permissions and the role hierarchy were examined, and data sharing rules were configured. The other items in the diagram, such as localization, schema builder and widgets, are available in the Settings menu of the application, but they were not configured in this experiment.

Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users. For a job application system, this is the area that would govern external applicants who access the application through a portal. No portal user permissions were configured in this experiment.

Role-Based Access and Data Sharing Flow

Figure 7 is a conceptual diagram of the security flow: User → User Permissions → Role → Data Sharing Rule → Access Level → Job Application Data. It represents how access is controlled in principle and is not evidence that each step was configured.

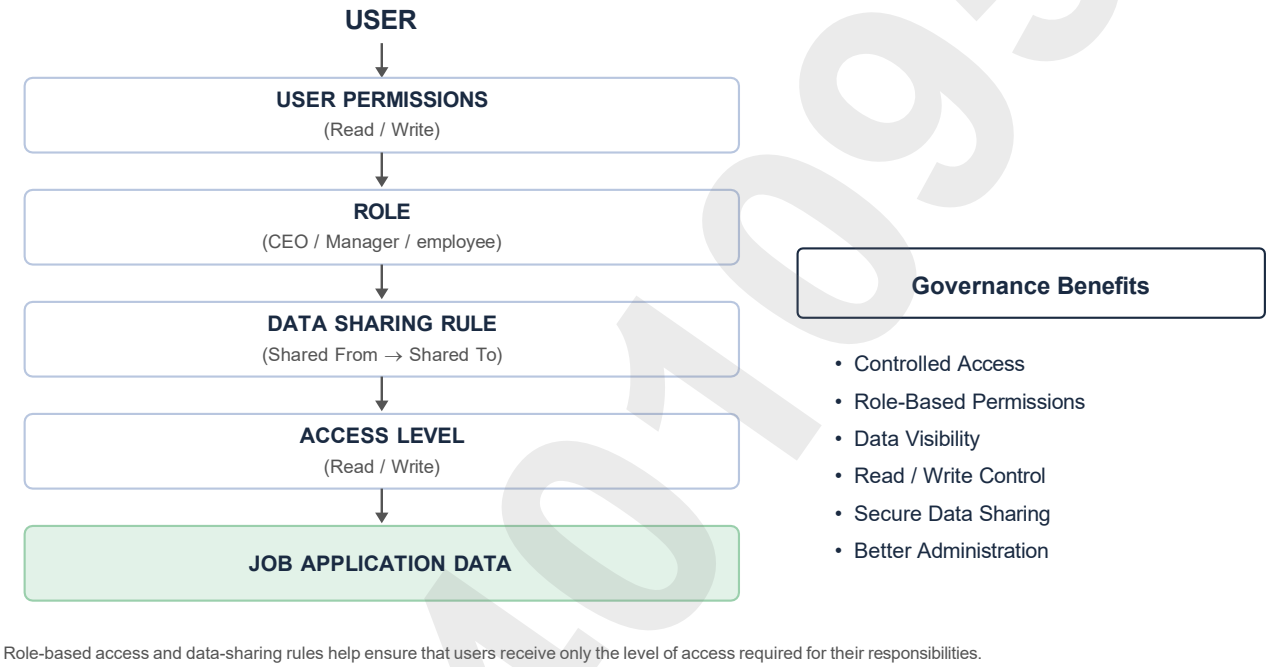


Figure 7: Role-Based Access and Data Sharing Flow (Conceptual Diagram)

- A user is first given permissions (Read / Write).
- The user belongs to a role, such as CEO, Manager or employee, which determines the position in the hierarchy.
- A data sharing rule specifies the records shared from one role or user to another.
- The access level (Read / Write) defines what the recipient can do, and this controls access to the Job Application data.

Governance Observations

- Three permission sets (manager, sadhana and Write) were listed in the Permissions tab.
- The Add Permission Set panel offers module-level permissions (Access, View, Edit, Delete, More) for the Job Application form and the dashboard.
- The role hierarchy contains the CEO role with the Manager and employee roles beneath it.
- Two data sharing rules were listed: sadhana (Manager to Manager, Read/Write) and revathy (employee to SADHANA, Read Only).
- The account was on a trial with an Upgrade option; no upgrade-only feature was configured.

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Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the permission sets manager, sadhana and Write were examined, the role hierarchy with the CEO role and the Manager and employee roles beneath it was viewed, and data sharing rules were configured to share the records of the Job Application form between roles and with the user SADHANA at the Read/Write and Read Only levels.

The experiment therefore shows the access-control tools that Zoho Creator offers: permission sets decide what a user can do, roles decide which records a user can see, and data sharing rules extend visibility with a chosen access level. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications.

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